

A Covid-19 Epidemiological Analysis and Forecasting Dashboard for Hospitals using Time-Series Analysis and Deep Learning

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Abstract—COVID-19 is a disease caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) that, to date, has over 774 million confirmed cases and claimed over 7 million lives. Forecasts are therefore essential as they aide in the necessary preparation of healthcare logistics. In this study, forecasting models such as ARIMA, RNN and LSTM were developed to predict Covid-19 time-series data relevant to hospitals, which were then evaluated and compared using MAE and RMSE. Across the different variables, LSTM had the least error. A web-based dashboard was also developed for forecasting these time-series variables for healthcare institutions, particularly those which are accommodating Covid-19 patients.

Index Terms—Covid-19, dashboard, time series forecasting, ARIMA, RNN, LSTM

I. INTRODUCTION

It was in the capital of China's Hubei province of Wuhan in December 2019 when the coronavirus disease 2019 (COVID-19), a condition caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), was first identified. In less than two months, it has swept across the globe. It was on the 30th of January, 2020, when the World Health Organization (WHO) declared the outbreak to be a Public Health Emergency of International Concern and on March 11, 2020, it was declared by the same as a pandemic [1], [2]. As of February 2024, there are over 774 million total cumulative reported cases and over 7 million deaths worldwide spanning over 230 countries and territories, with the Philippines having a total of 4.1 million confirmed cases and over 66.9 thousand deaths [3].

The COVID-19 pandemic caused unprecedented stress on health care systems around the globe as it required treatment capabilities and resources that exceeded the expected emergency surge capacities. Naturally, this had a domino effect as there was a resulting shortage in personal protective equipment (PPE), lack of isolation spaces, critical resource supply chains were strained and emergency rooms were overwhelmed [4]. Unfortunately, there were cases of deaths due to shortages of ventilators and dialysis machines which could have been avoided if hospitals had enough critical equipment and personnel to meet the needs of the patients. Though availability of funds to purchase such

resources is definitely essential, accurate information on the demand as well as projections on the number and timing of cases requiring various types of treatment are just as necessary. Several tools were developed to assist hospital administrators, physicians, emergency response personnel, and governmental agencies to aide in planning and responding to Covid-19 [4], which include information dashboards.

The International Severe Acute Respiratory and emerging Infection Consortium (ISARIC) is a consortium based in the United Kingdom that supplies clinical research network around the world and provides agile research response to outbreak-prone infectious diseases, such as Covid-19 [5]. Tools have been developed by the ISARIC for those in the medical field to collect, investigate, and store data in a standardized manner. At present, ISARIC already has three active hubs: in Africa, South Asia, and South America. The hub in Southeast Asia is currently lead by the Philippines, specifically by the Tropical Disease Foundation and Makati Medical Center. The development is still in the early stages and among the problems in its implementation is the lack of a system for Covid-19 dashboard. Hence, this study aims to develop a Covid-19 dashboard that may spearhead the successful launch of the ISARIC Hub in the Philippines [5].

The ISARIC dashboard provides various visualizations about the present state of Covid-19 based on the data provided to it [5]. However, it lacks visualization about forecasting of the future state of the disease, most specifically, in terms of the number of Covid-19 cases and some of its related metrics.

The main objective of this study is to create a web-based system based on ISARIC Covid-19 clinical dashboard that can facilitate visual representation and forecasting the number of Covid-19 cases and some of its related metrics based from the provided electronic health records. A forecasting page for the Covid-19 clinical dashboard will be a valuable tool in monitoring the current status of Covid-19 as well as anticipate the next condition in a community. Policymakers can use this tool in order to adjust their policies based on what is currently needed to control the spread of the virus and to maintain low transmission. Healthcare institutions and

hospital administrators can also use this tool to anticipate surges and predict spikes of Covid-19 cases and accordingly prepare the necessary logistics.

II. RELATED WORK

During the pandemic, dashboards have become essential tools as they provide real-time, summative information as well as visualizations that provide important insights necessary for planning and decision-making. The systematic review studies done by Ivankovic et. al. [6] as well as that by Vahedi et. al. [7] pointed out features constituting actionable COVID-19 dashboards. They noted that good-quality dashboards are important tools that aid policymaking, coordination, and coherence in implementation, centralized information, organization, continuous monitoring, and management of the Covid-19 crisis. The study [6] conducted systematic review in 2021 was based on the PRISMA Statement [8]. It was found that all interactive Covid-19 dashboards mainly used graphs and geographical maps, as well as charts that helped in providing quality and accuracy of the data presentation. Across the world, different types of dashboards were used to counter the rapid spread of Covid-19 worldwide, maximizing the advantages brought by the information technology and advanced computer sciences. Aside from disease monitoring, tacking of infected areas, and reporting of Covid-19 prevalence on a specific scope, interactive Covid-19 dashboards also aided in tracking diagnosis and monitoring of a population at large, and proper resource management for hospitals and other higher levels of medical service. The unprecedented increase in Covid-19 cases worldwide also brought the need for an effective model for prediction of future Covid-19 cases [9]. There were already studies done that provided models for the prediction of the number of Covid-19 cases, the most notable of which is the use of time series modeling. Time series modeling uses historical data to forecast expected outcome in a future time. In a study done by Somyanonthanakul et al. [10], the Autoregressive Integrated Moving Average (ARIMA), a time series modeling approach, was used to analyze the number of Covid-19 patients. In addition to that, an association rule mining (ARM) was also done in order to support the analysis of ARIMA. They were able to predict the daily hospital admission of Covid-19 patients with confidence intervals between 5% and 95%. A similar approach was done by Tyagi et al. [11], wherein it was attempted to predict the number of Covid-19 cases, as well as the requirement of ICU beds, ventilators, and isolation beds using ARIMA model. Singh et. al [12] explored 7 models in their time series analysis of Covid-19 cases. It has been observed that the ARIMA model stood out in terms of forecasting accuracy. On the other hand, the study of Alassafi et al. [13] used other machine learning techniques, specifically deep learning techniques, in predicting Covid-19 new cases and new deaths. They used recurrent neural network and long short-term memory network in their prediction. The former garnered 93.45% accuracy while

the latter achieved 98.58% accuracy. Dashboards are indeed helpful in humanity's combat against the Covid-19 pandemic by providing a powerful tool for public healthcare planning in the new normal.

III. METHODOLOGY

The main goal of this study is to create web-based dashboard for forecasting Covid-19 metrics relevant to hospitals, particularly newly admitted count, newly positive count, daily active cases count, daily deaths, and daily bed demand.

A. The System

The web-based dashboard summarizes and presents Covid-19 forecasts for the public to view. The data is provided by personnel from the healthcare field, particularly researchers. The models for graphs, forecasts, and reports are provided by the developer. Public users can view graphs, see forecasts, and download reports from the dashboard.

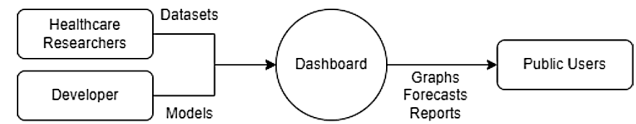


Fig. 1: Context Diagram of the System.

B. The Dataset

The data used in this study are from the joint database of Covid-19 electronic health records of the Makati Medical Center and Tropical Disease Foundation. Up to the date of this study, data have been entered for 2356 individuals from various sites across the Philippines for whom data collection have commenced on or before 15 March 2021.

C. The Block Diagram

Time series data is pulled from the dataset prior to being subject to data cleaning techniques, such as the removal of invalid values, and data imputation.

Afterwards, stationarity tests were done to see if the time series is stationary in nature. Differencing, lag transformation, and normalization of the time series data were done in order to facilitate the efficiency and accuracy of the model during the training phase. 70% of the data was classified as the train set, and the remaining 30% as the test set. Since, this deals with time-stamped data, the test set is composed of all the most recent data. The train set is used to train the autoregressive integrated moving average (ARIMA), recurrent neural networks (RNN), and long short term memory (LSTM) forecasting models. The models were then used to forecast more recent values and evaluated by comparing their respective forecasts with the test set using following metrics: 1) mean absolute error (MAE) and 2) root mean square error (RMSE)

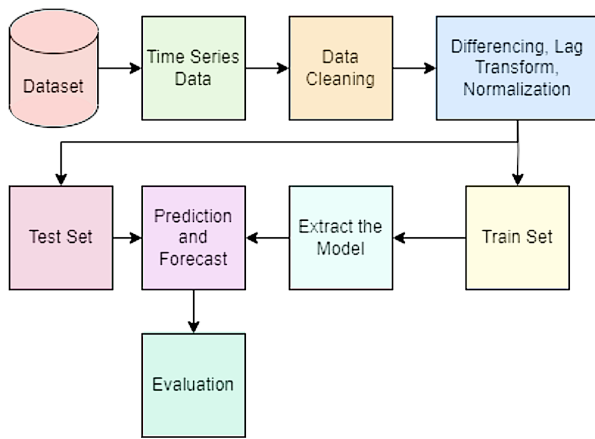


Fig. 2: The Block Diagram for Forecasting

IV. RESULTS AND DISCUSSION

In this study, ARIMA, RNN and LSTM forecasting models were developed to predict Covid-19 time-series data relevant to hospitals, particularly newly admitted count, newly positive count, daily active cases count, daily deaths, and daily bed demand.

A. Newly Admitted Cases Count

The time series data of newly admitted cases count from the data set was subjected to Augmented Dickey-Fuller Test (ADF) and Partial Autocorrelation Function (PCF). The results of the ADF showed that this time series data is stationary and the PCF showed that a significant lag is seen at lag 1. The most optimal ARIMA Model trained from the newly admitted time series data was ARIMA(2,0,1).

In training the RNN Model, the time series data was subjected to lag transform, and normalization. No differencing was applied on the data because it was already stationary. The resulting RNN model was composed of 675 neurons with 25 layers and a dense layer with 26 neurons, with a total of 701 neurons.

The LSTM model was also trained with the same pre-training approach as the RNN model. The resulting model was composed of 10400 neurons with 50 layers and a dense layer of 51 neurons, with a total of 10451 neurons.

For comparison, the forecasts of the three models for the newly admitted cases time series data is shown in Figure 3.

B. Newly Positive Cases Count

For the daily count of newly positive patients, the ADF test showed that the data is not stationary and the PCF test showed significance at lag 1. Since, the data is not stationary, differencing was necessary prior to the training in order for it to become stationary. The most optimal ARIMA model trained for the forecasting of the count of newly positive data was ARIMA(2,1,2).

In training the RNN and LSTM model for this data, differencing, lag transformation, and normalization was applied.

The resulting RNN model was composed of 675 neurons with 25 layers and a dense layer with 26 neurons, with a total of 701 neurons. The LSTM model, on the other hand, was also trained with the same pre-training approach as the RNN model. The resulting model was composed of 10400 neurons with 50 layers and a dense layer of 51 neurons, with a total of 10451 neurons. LSTM is the best performing model and the forecast is shown in Figure 4.

C. Daily Active Cases Count

The data on daily active cases count was found to be not stationary according to the ADF test and the PCF test shown significance at lag 1. The most optimal ARIMA model trained for the forecasting of the count of daily active cases data was ARIMA(1,1,1).

In training the RNN and LSTM model for this data, differencing, lag transformation, and normalization was applied. The resulting RNN model was composed of 675 neurons with 25 layers and a dense layer with 26 neurons, with a total of 701 neurons. The LSTM model was also trained with the same pre-training approach as the RNN model. The resulting model was composed of 10400 neurons with 50 layers and a dense layer of 51 neurons, with a total of 10451 neurons. LSTM is the best performing model and the forecast is shown in Figure 5.

D. Daily Deaths Count

The data for deaths was found to be not stationary by ADF Test and the optimal lag was not able to be produced by PCF. The most optimal ARIMA model trained for the forecasting of the count of daily deaths data was ARIMA(0,0,1).

In training the RNN and LSTM model for this data, differencing, lag transformation, and normalization was applied. The resulting RNN model was composed of 675 neurons with 25 layers and a dense layer with 26 neurons, with a total of 701 neurons. The LSTM model was also trained with the same pre-training approach as the RNN model. The resulting model was composed of 10400 neurons with 50 layers and a dense layer of 51 neurons, with a total of 10451 neurons. The LSTM model was also trained with the same pre-training approach as the RNN model. The resulting model was composed of 10400 neurons with 50 layers and a dense layer of 51 neurons, with a total of 10451 neurons. RNN is the best performing model and the forecast is shown in Figure 6.

E. Daily Bed Demand

The data for daily bed demand was shown to be stationary according to ADF test. Moreover, PCF showed that the significant lag is at 1. The most optimal ARIMA model trained for the forecasting of the count of daily bed demand data was ARIMA(2,1,2). In training the RNN and LSTM model for the daily bed demand data, differencing, lag transformation, and normalization was applied. The resulting RNN model was composed of 675 neurons with 25 layers and a dense layer with 26 neurons, with a total of 701 neurons. LSTM

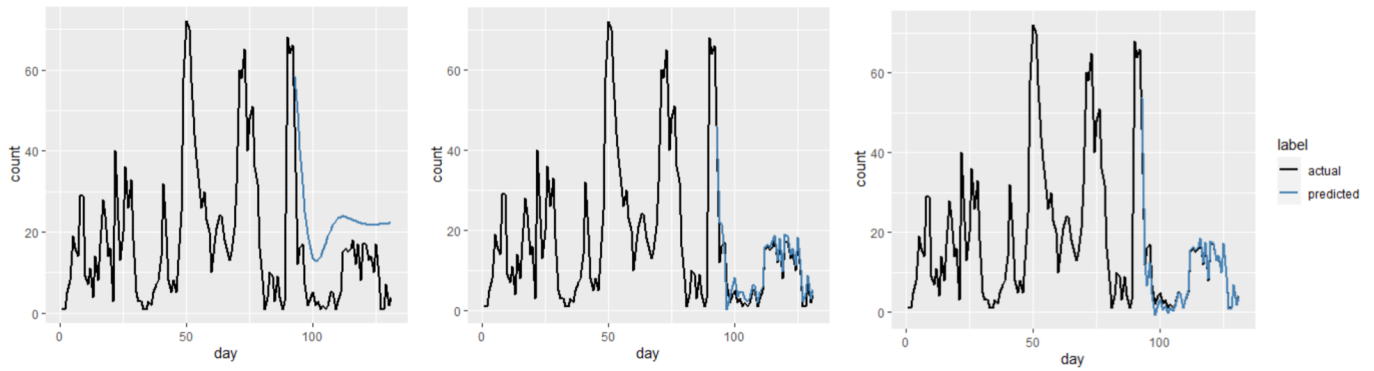


Fig. 3: Newly Admitted Cases forecast of the ARIMA, RNN and LSTM models, respectively.

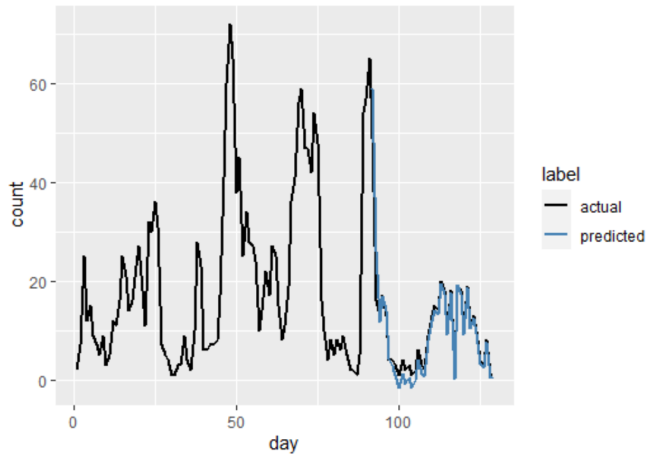


Fig. 4: LSTM Forecast for Newly Positives Time Series Data

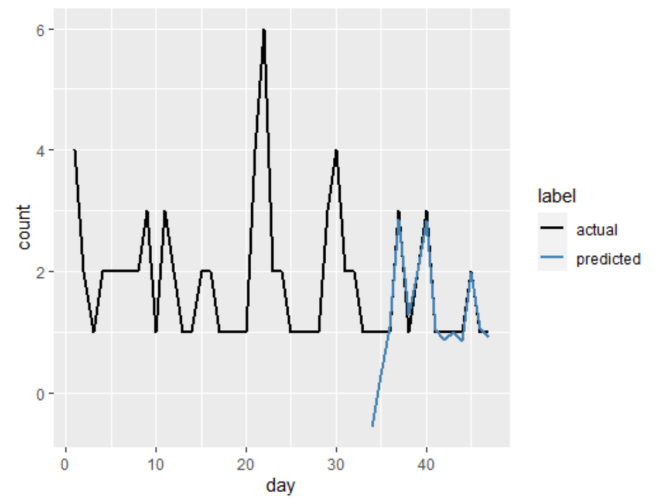


Fig. 6: RNN Forecast on Daily Deaths Time Series Data

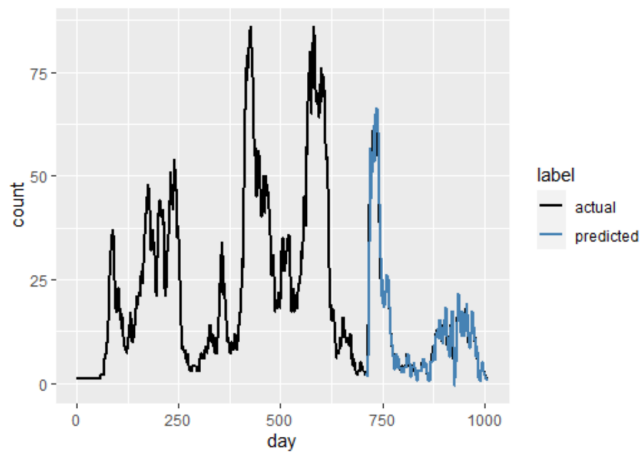


Fig. 5: LSTM Forecast on Daily Active Time Series Data

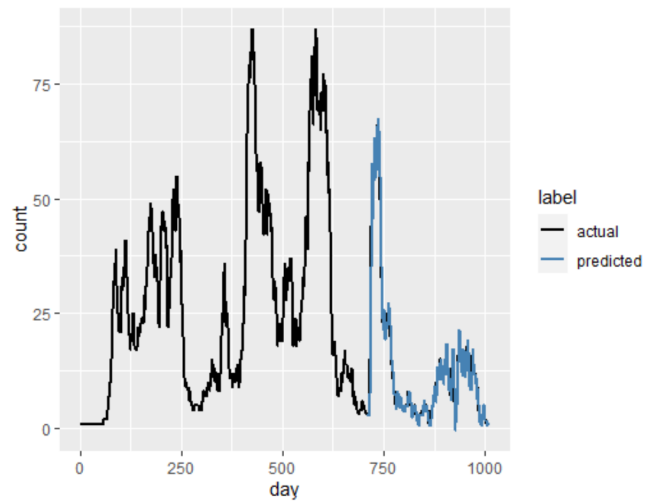


Fig. 7: LSTM Forecast on Daily Bed Demand Time Series Data

is the best performing model and the forecast is shown in Figure 7.

Table 1 shows the performance of ARIMA, RNN and LSTM on forecasting Newly Admitted Cases, Newly Posi-

NEWLY ADMITTED CASES						
Models	ARIMA		RNN		LSTM	
Metric	MAE	RMSE	MAE	RMSE	MAE	RMSE
Train	7.9437	11.4201	0.0331	0.0021	0.0210	0.0012
Test	13.7906	15.29631	1.5186	2.8224	1.5808	2.1169
NEWLY POSITIVE CASES						
Models	ARIMA		RNN		LSTM	
Metric	MAE	RMSE	MAE	RMSE	MAE	RMSE
Train	7.3575	10.1305	0.0015	0.0387	0.0152	0.12329
Test	12.7106	14.1477	1.9870	2.8194	1.5789	1.2476
DAILY ACTIVE CASES						
Models	ARIMA		RNN		LSTM	
Metric	MAE	RMSE	MAE	RMSE	MAE	RMSE
Train	1.7097	2.5548	0.0483	0.2198	0.0415	0.2037
Test	11.7676	18.6311	1.8861	2.6237	1.5793	2.6667
DAILY DEATHS						
Models	ARIMA		RNN		LSTM	
Metric	MAE	RMSE	MAE	RMSE	MAE	RMSE
Train	0.8561	1.0809	0.0823	0.2869	0.1272	0.3567
Test	0.8311	0.9037	0.9872	1.1475	1.1215	1.1762
DAILY BED DEMAND						
Models	ARIMA		RNN		LSTM	
Metric	MAE	RMSE	MAE	RMSE	MAE	RMSE
Train	1.5088	2.2767	0.0450	0.2121	0.0165	0.1285
Test	6.3258	7.8955	0.7896	1.2479	0.1470	1.1475

TABLE I: Performance of ARIMA, RNN and LSTM on forecasting Newly Admitted Cases, Newly Positive Cases, Daily Active Cases, Daily Deaths and Daily Bed Demand

tive Cases, Daily Active Cases, Daily Deaths and Daily Bed Demand based on MSE and RMSE.

Clearly, deep learning algorithms such as recurrent neural networks and long short term memory were found to be able to forecast the number of Covid-19 new cases, active cases, deaths, and even demand on beds, as well as the rise and peaks of these numbers in the time series data.

The LSTM visualization, along with those of the other models were featured in the web-based system, a screenshot of which is shown in Figure 8. The ARIMA projections are shown here for the 90% and 95% confidence intervals.

We thus see this dashboard as being a very useful tool for healthcare institutions, particularly those which are accommodating Covid-19 patients.

V. CONCLUSION

In this study, ARIMA, RNN and LSTM forecasting models were developed to predict Covid-19 time-series data relevant to hospitals, particularly newly admitted count, newly positive count, daily active cases count, daily deaths, and daily bed demand. A web-based dashboard featuring visualizations of these models was also developed. It is hoped that this tool will be found very useful for healthcare institutions, particularly those which are accommodating Covid-19 patients. Future developers can improve this dashboard by seeking for other relevant statistical visualizations.

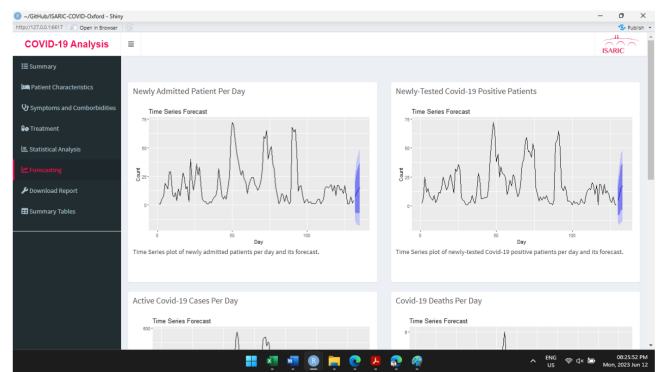


Fig. 8: LSTM Forecast on Daily Bed Demand Time Series Data

As machine learning and deep learning algorithms improve, the forecasting functionality also continue to improve. Other interfaces may also be developed so that this dashboard can be able to connect with other systems for the sake of data transfer, system collaboration, and the like.

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